

Introduction

Poly(lactic acid) (PLA) is a widely used biodegradable plastic in **food storage containers**, but it currently lacks **antimicrobial properties**, making it susceptible to **mold growth** and **bacterial contamination**. As a result, fruits and vegetables stored in PLA containers are more likely to spoil, contributing to a major worldwide problem of **food waste**.

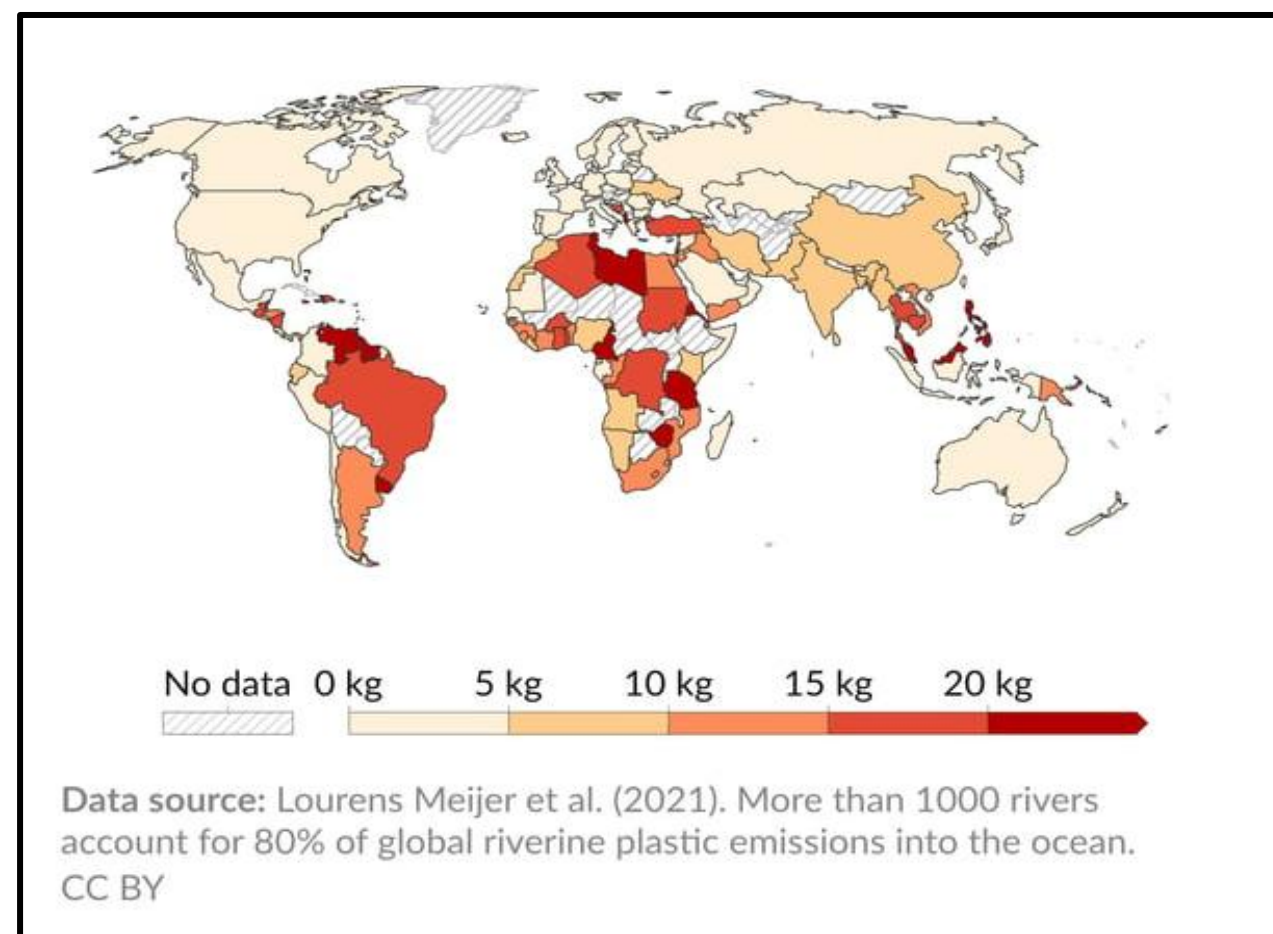


Figure 1. World Map of Mismanaged Plastic Waste per Person

Poor food preservation also poses a **public health concern**, as microbial growth on stored food increases the risk of **foodborne illness**. While antimicrobial coatings for food packaging exist, many are expensive, have chemical concerns and negative environmental impacts. There is now a need for a **safe, natural, and environmentally safe solution** that improves the antimicrobial performance of PLA containers. Developing a surface coating for PLA can help improve **food safety, extend shelf life, and reduce food waste**. By addressing both **food safety** and **environmental concerns**, advancements in antimicrobial surface coatings can help **sustainability** in **effective food preservation**.

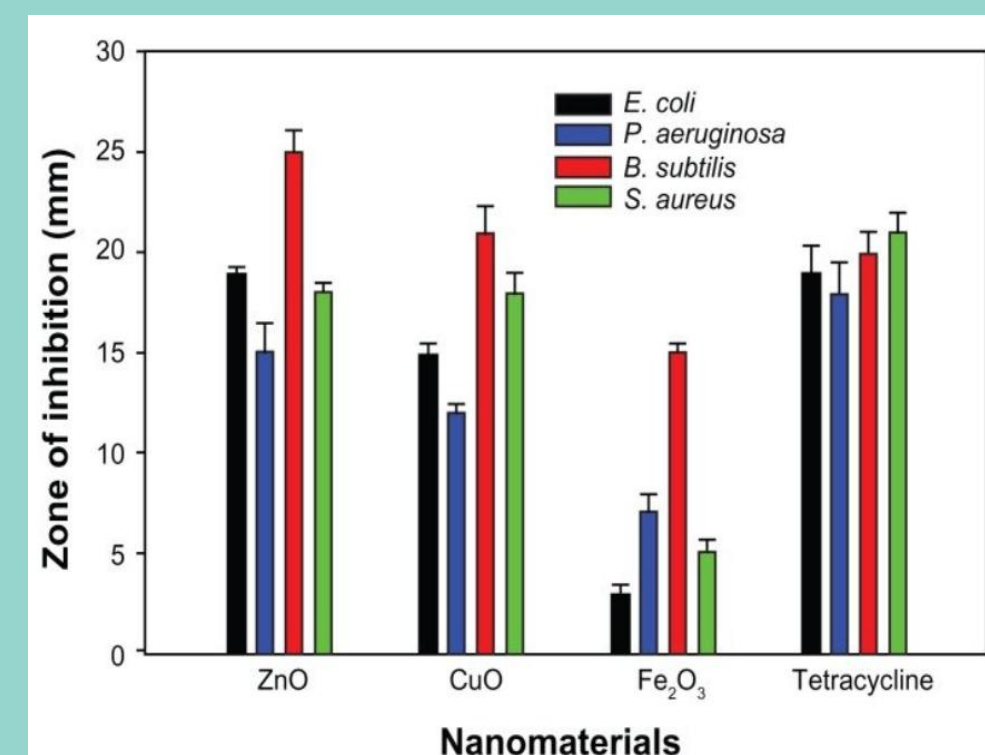


Figure 2. Graph showing zone of inhibition from different metal oxides against microorganisms

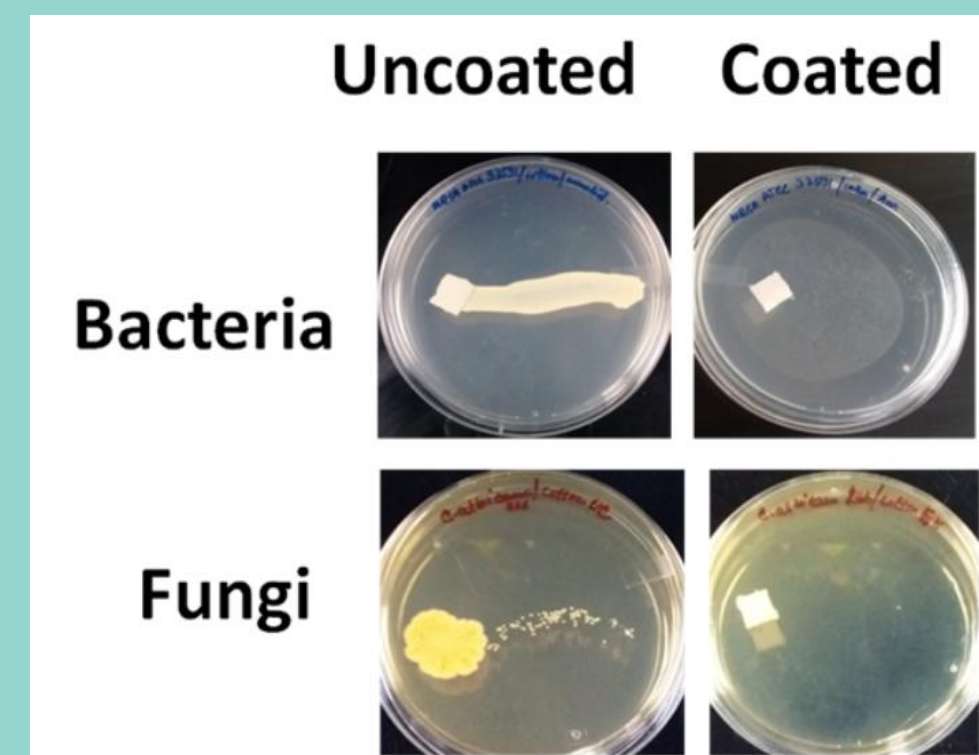


Figure 3. Bacterial and Fungal Growth on Uncoated Compared to Antimicrobial Coatings

Purpose

Surface-coated PLA food storage containers with **enhanced antimicrobial activity** were **designed** and **evaluated** using **lactic acid** and **zinc oxide (ZnO)**. Rather than modifying the PLA fiber, a **brush-applied** surface coating was used as a **cost effective** and **practical** solution for **real world applications** in the future. Effectiveness was tested based on **visual changes** over a week, such as **mold growth, bacterial presence, and shelf life** on containers filled with **strawberries**. Surface coatings have the potential to improve **food safety**, reduce **spoilage**, and support more **sustainable** food packaging options. Enhancing antimicrobial properties may help extend shelf life while reduce overall **food waste**.

Utilizing a Zinc Oxide and Lactic Acid Coating to Reduce Mold and Bacterial Growth on PLA Food Storage Containers

Methods

Coating Creation

- Prepared a PVA binder solution by dissolving 6g of PVA in 200 mL distilled water on a hotplate at 120 degrees C
- Dispersed 2g ZnO nanoparticles and 2g lactic acid into the binder solution to form a uniform coating solution.



Figure 4. PVA Binder Solution Before Heating

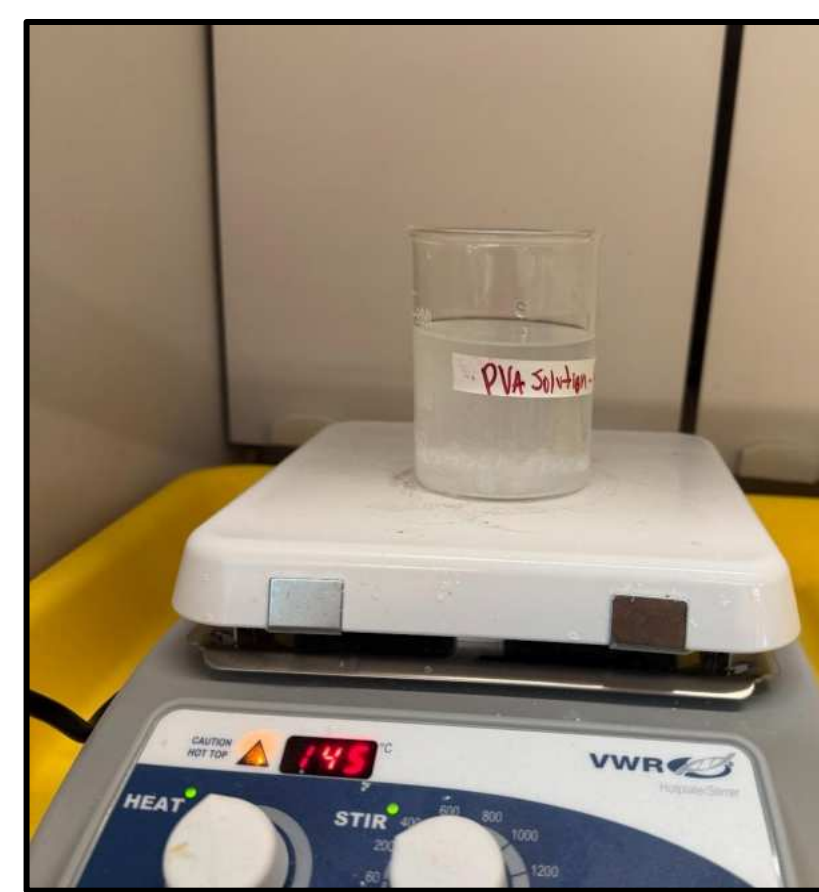


Figure 5. Final Solution on Hotplate

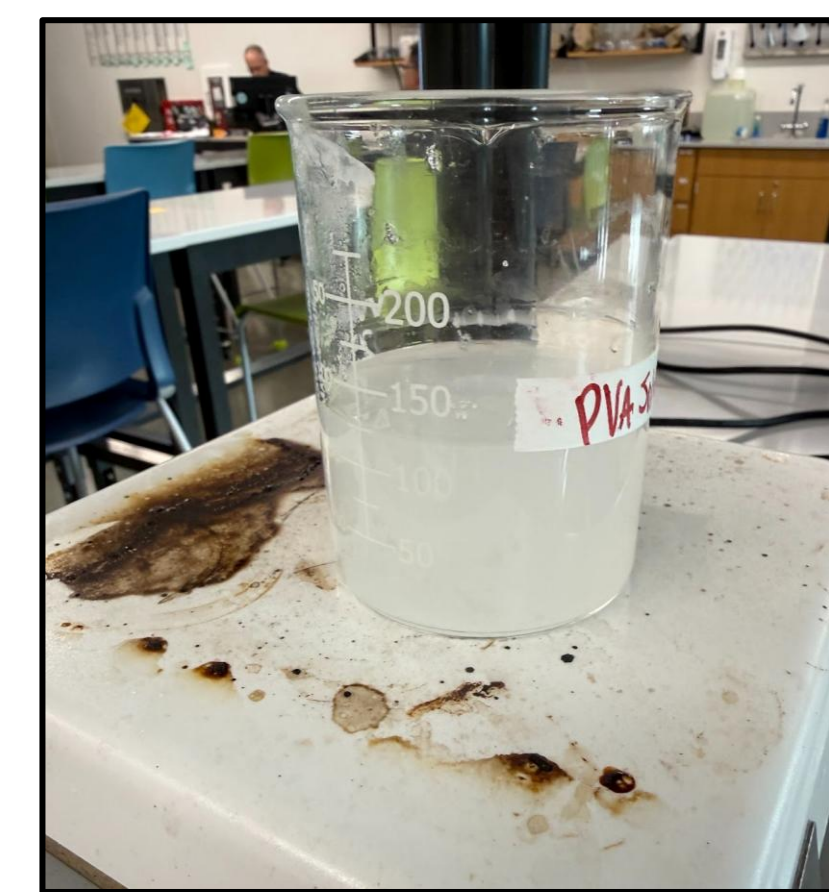


Figure 6. Final Solution Fully Dispersed

Coating Containers

- Applied coating onto PLA containers using kitchen pastry brush with ~5-10 mL of solution per container
- Dried for 5 minutes before adding second layer of coating
- Cured coated containers for 1-2 days in room temperature



Figure 7. Brush-coated PLA Container



Figure 8. Completed Brush-coated PLA Containers



Figure 9. Second Trial - Completed Brush-coated PLA Containers

Testing Containers

- Filled 4 coated and 4 uncoated containers with equal amount of strawberries
- Left containers untouched for 1 week while checking every ~2 days documenting any physical changes
- Compared results between uncoated and coated containers

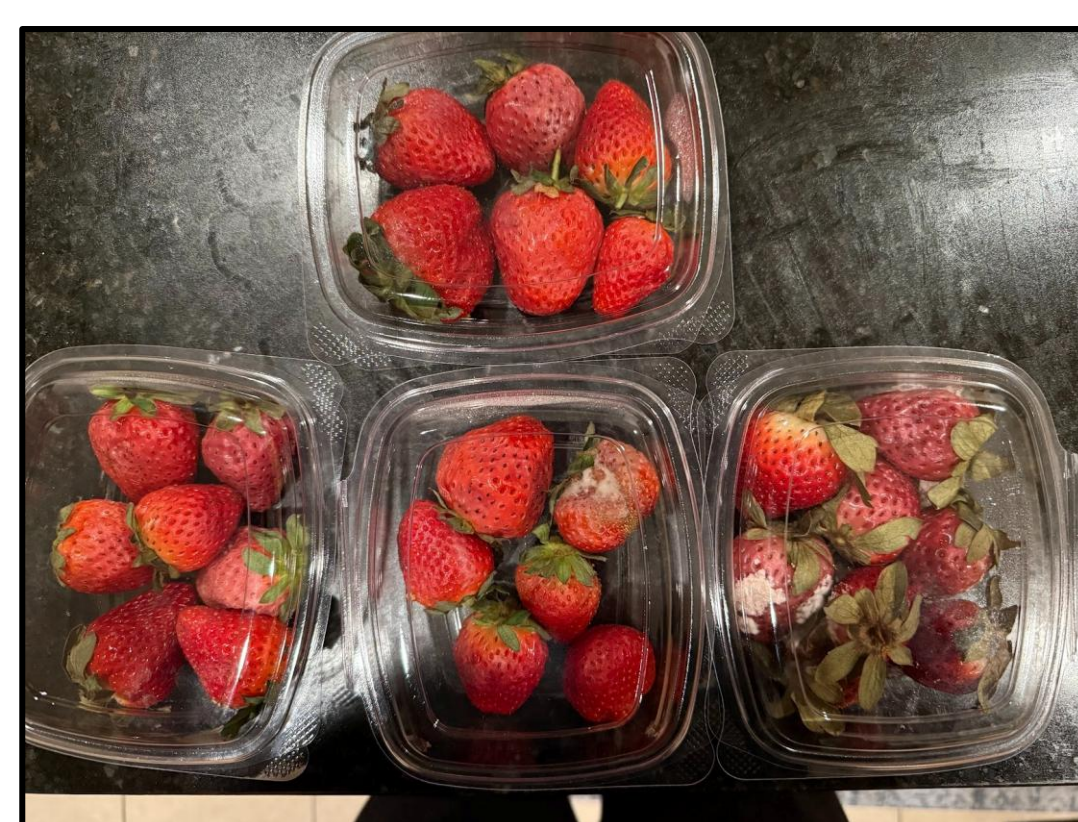


Figure 10. Uncoated Container Final Day of Testing



Figure 11. Coated Container Final Day of Testing

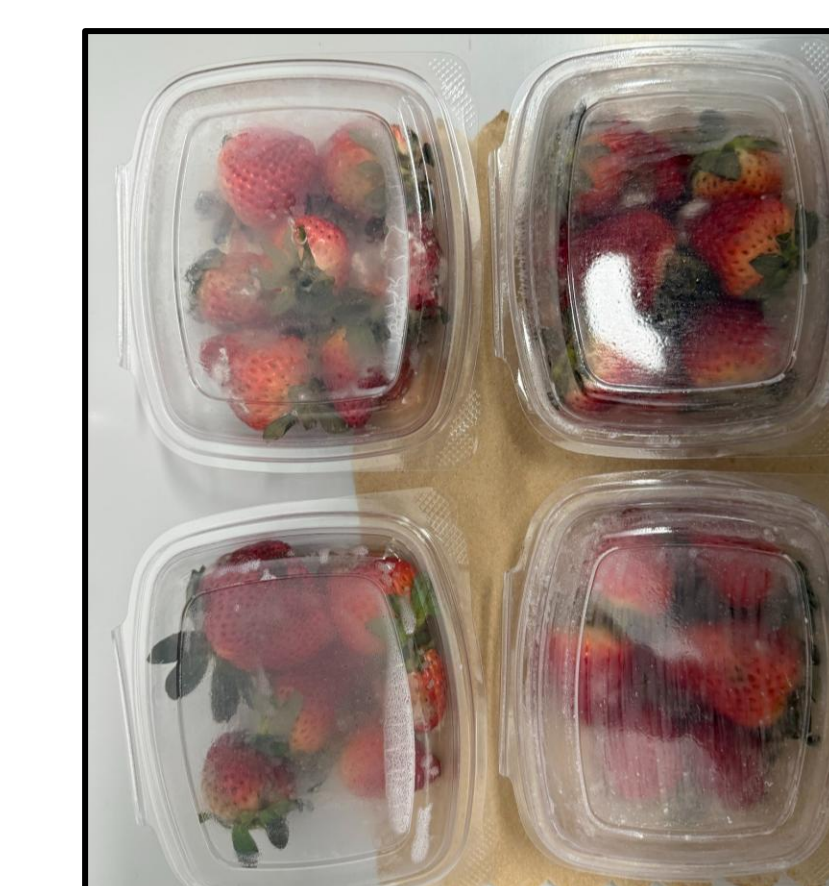


Figure 12. Second Trial - Coated Container Final Day of Testing

Results

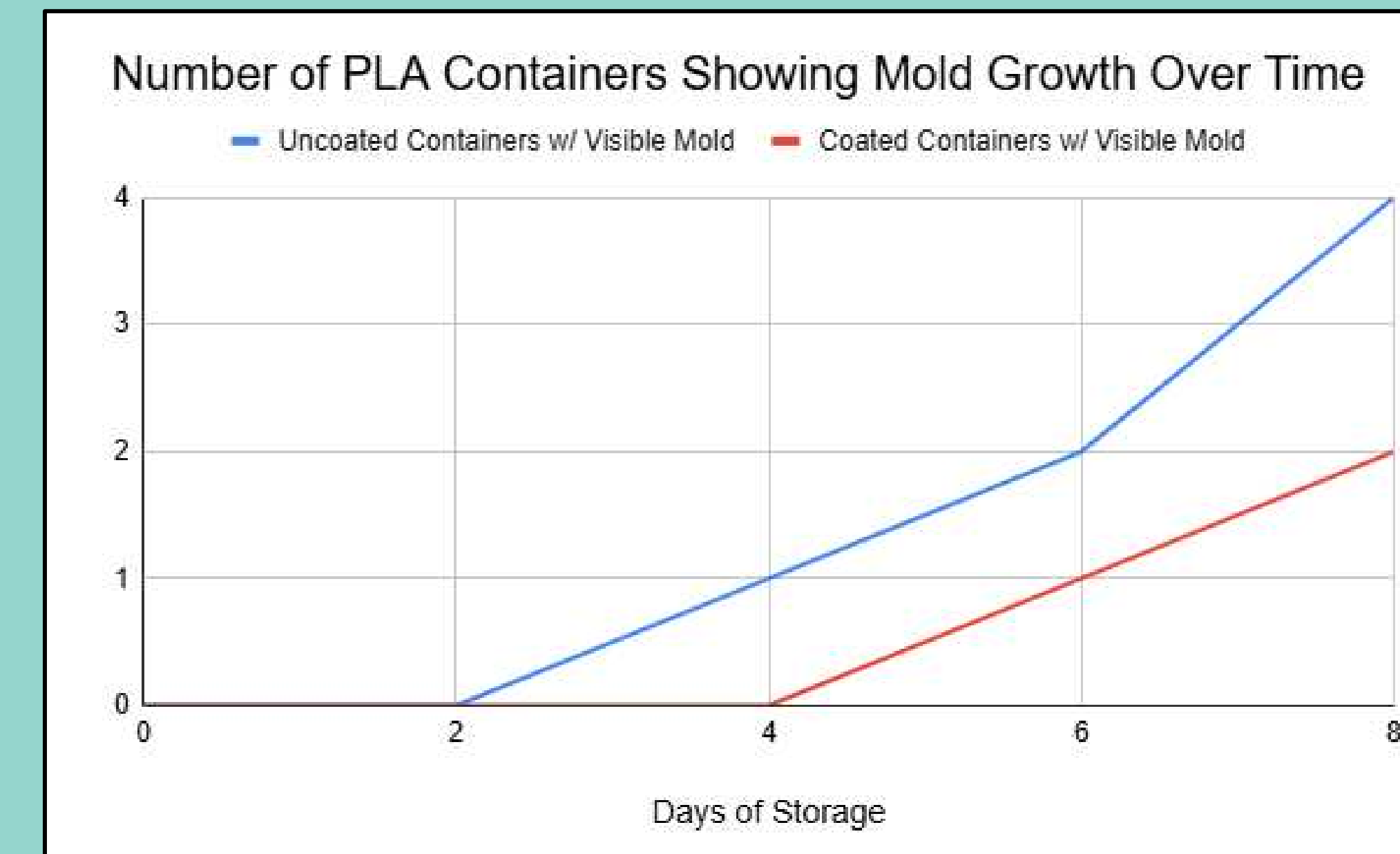


Figure 13. Coated vs Uncoated Containers Showing Mold Growth Over 8 Days Showing Delayed Visible Mold Growth in Coated Containers

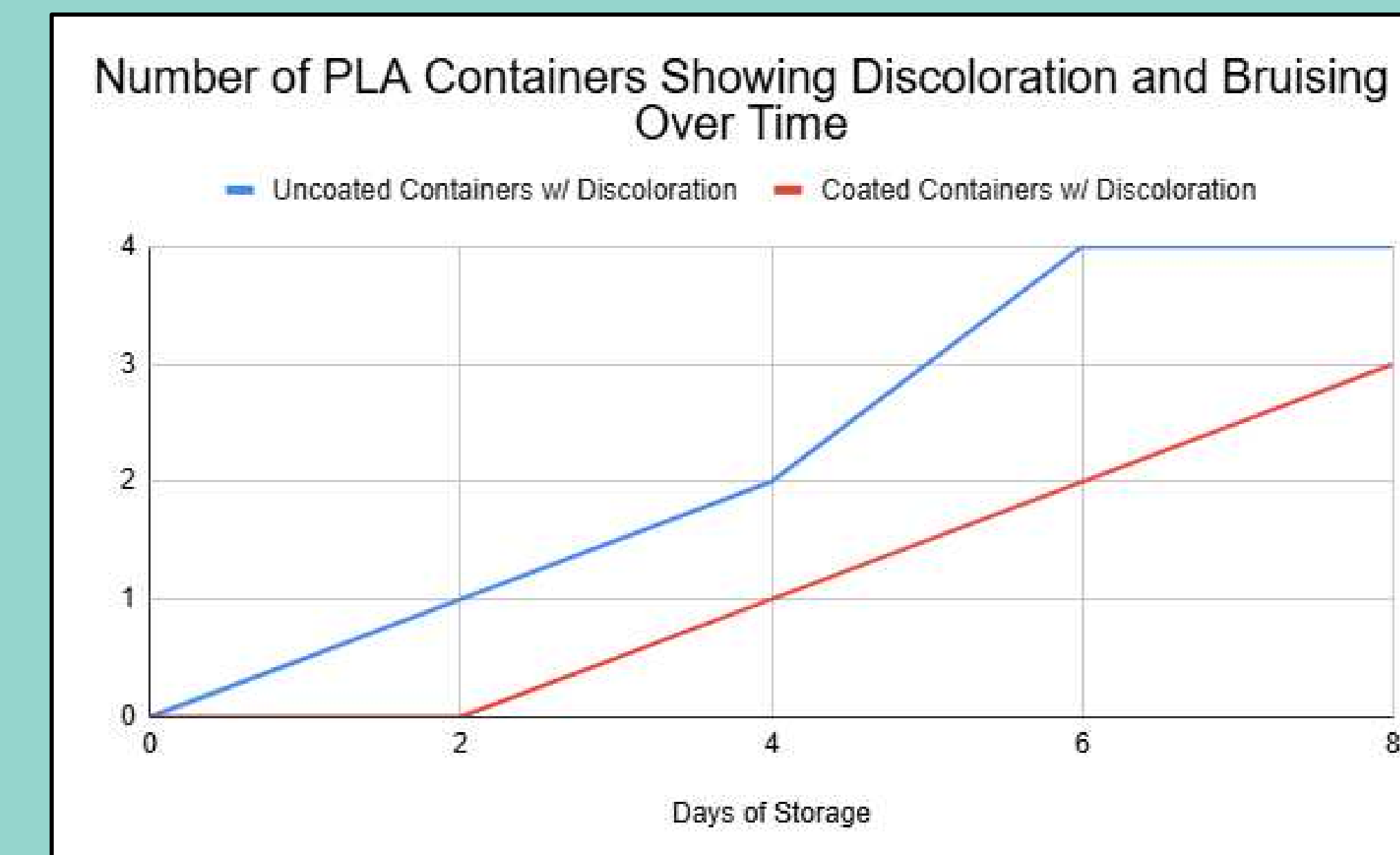


Figure 14. Coated vs Uncoated Containers Showing Discoloration and Bruising Over 8 Days Showing Delayed Visible Bruising

Conclusion

The antimicrobial coating of PVA, ZnO, and lactic acid showed **effectiveness** in reducing mold growth and spoilage compared to uncoated PLA containers. While the coated containers showed **slight visual differences**, the overall results from the trial **did not significantly improve** food preservation in the conditions of testing. These findings show that while antimicrobial coatings have **potential**, this specific formulation and application method used in this study may require **further research** and **optimization**.

Future research should include multiple trials to improve reliability and allow for better analysis. The coating formulation could be adjusted by changing the concentration of ZnO or lactic acid to enhance antimicrobial performance. Also, adding quantitative data on the true mold percentage could be collected to further improve future analyses.

References

- Duan, L., Yan, F., Zhang, L. (2022, November 28). ZnO@Poly(vinyl Alcohol)/Poly(lactic acid) Nanocomposite Films for the Extended Shelf Life of Pork by Efficient Antibacterial Adhesion. ACS Omega. <https://pubs.acs.org/doi/10.1021/acsomega.2c03016>
- Kim, I., Viswanathan, K., Kasi, S., Sadeghi, K., & Seo, J. (2020). ZnO Nanostructures in Active Antibacterial Food Packaging: Preparation Methods, Antimicrobial Mechanisms, Safety Issues, Future Prospects, and Challenges. Food Reviews International, 38(4), 537–565. <https://doi.org/10.1080/87559129.2020.1737709>
- Rajendran, D. S., Venkataraman, S., Jha, S. K., Chakrabarty, D., & Kumar, V. V. (2024, March 12). A review on bio-based polymer poly(lactic acid) potential on sustainable food packaging. Food science and biotechnology. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11091039/>